

Why Does the Inequality Bind? (Understanding $\lambda_1 = p_1 - p_2$)

ECO 310 - The Deeper Question

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1 The Question

We have: $-p_1 + \lambda_1 + \lambda_3 \leq 0$

After substituting $\lambda_3 = p_2$, this becomes: $\lambda_1 \leq p_1 - p_2$

Why do we assume this holds with **equality**: $\lambda_1 = p_1 - p_2$?

2 Short Answer

Quick Answer

Actually, we don't always assume it binds!

The condition $\lambda_1 \leq p_1 - p_2$ tells us the **maximum possible value** of λ_1 .

For perfect substitutes specifically, it turns out that λ_1 takes its maximum value at the optimum. But this is a result we derive, not something we assume.

3 The Full Story

3.1 What KKT Conditions Tell Us

The Karush-Kuhn-Tucker (KKT) optimality conditions for our problem include:

$$\frac{\partial \mathcal{L}}{\partial x_1} \leq 0, \quad x_1 \geq 0, \quad x_1 \cdot \frac{\partial \mathcal{L}}{\partial x_1} = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial x_2} \leq 0, \quad x_2 \geq 0, \quad x_2 \cdot \frac{\partial \mathcal{L}}{\partial x_2} = 0 \quad (2)$$

These conditions are **necessary** for optimality (must be true at any optimal solution).

3.2 What We Actually Know

From the guess that $x_1 = 0$ and $x_2 > 0$:

We know for certain:

- $\lambda_2 = 0$ (from complementary slackness with $x_2 > 0$)
- $\lambda_3 = p_2$ (from the x_2 FOC being exactly zero)

- $x_2 = u^*$ (from utility constraint with $x_1 = 0$)

We know only a constraint on λ_1 :

- $\lambda_1 \leq p_1 - p_2$ (from the x_1 FOC inequality)
- $\lambda_1 \geq 0$ (all Lagrange multipliers must be non-negative)

So $\lambda_1 \in [0, p_1 - p_2]$ is feasible.

4 Why We Set $\lambda_1 = p_1 - p_2$

There are a few ways to think about this:

4.1 Approach 1: It's the Unique Value That Works

For perfect substitutes, there's a unique optimal solution. If the optimal λ_1 were strictly less than $p_1 - p_2$, this would suggest the solution is not fully determined by the first-order conditions alone, which contradicts the uniqueness of the corner solution.

Intuition

Think of it this way: the system of optimality conditions should uniquely determine all the variables.

If λ_1 could be anywhere in $[0, p_1 - p_2]$, we'd have infinitely many "solutions," which doesn't make sense for a unique optimization problem.

The fact that we're at a corner solution (buying only one good) suggests the constraints are "tight" - there's no wiggle room.

4.2 Approach 2: Economic Interpretation

Remember what λ_1 means: the shadow price of the constraint $x_1 \geq 0$.

Question: How much would your cost decrease if you could buy $x_1 = -\epsilon$ (slightly negative)?

Answer:

- You'd "sell" ϵ units of good 1 and get $p_1\epsilon$ dollars back
- You'd need to buy ϵ more of good 2 to maintain utility, costing $p_2\epsilon$
- Net savings: $(p_1 - p_2)\epsilon$ per unit

So the shadow price should be exactly $p_1 - p_2$, not something smaller!

4.3 Approach 3: The Math is Actually More Subtle

Here's the truth: we don't *assume* it binds. Instead:

1. We **guess** that $x_1 = 0, x_2 = u^*$ is optimal
2. We derive that IF this is optimal, THEN $\lambda_1 = p_1 - p_2$
3. We check that this solution satisfies ALL the KKT conditions
4. If it does, we've found an optimal solution

5 The Detailed Argument

Let me show you why $\lambda_1 = p_1 - p_2$ is the "right" value.

5.1 Setting Up the Problem

We want to minimize $-\mathcal{L} = p_1x_1 + p_2x_2$ (the actual cost) subject to all constraints.

The Lagrangian with Convention 2 is:

$$\mathcal{L} = -p_1x_1 - p_2x_2 + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(x_1 + x_2 - u^*)$$

At a maximum of $\mathcal{L}$ (which corresponds to minimum cost):

- We can't increase $\mathcal{L}$ by changing x_2 (since $x_2 > 0$, the derivative must be exactly 0)
- We can't increase $\mathcal{L}$ by increasing x_1 from 0 (derivative must be ≤ 0)
- We can't increase $\mathcal{L}$ by changing λ_3 (derivative must be 0)

5.2 Why the Inequality is Actually Tight

Consider what happens if $\lambda_1 < p_1 - p_2$ (strict inequality).

Then: $-p_1 + \lambda_1 + \lambda_3 < 0$

This means $\frac{\partial \mathcal{L}}{\partial x_1} < 0$ (strictly negative).

Interpretation: Increasing x_1 would *decrease* the Lagrangian.

But wait - we're at $x_1 = 0$. We can't decrease x_1 further (nonnegativity constraint).

However, the strict inequality $\frac{\partial \mathcal{L}}{\partial x_1} < 0$ means:

"If we could buy less than zero of good 1, the Lagrangian would increase (equivalently, cost would decrease)."

The shadow price λ_1 measures exactly how much the Lagrangian would increase per unit if we could violate $x_1 \geq 0$.

From our economic argument: violating $x_1 \geq 0$ by one unit (setting $x_1 = -1$) and adjusting x_2 to maintain utility would decrease cost by exactly $p_1 - p_2$.

Therefore: $\lambda_1 = p_1 - p_2$ (the full amount, not something smaller).

6 What Happens in Other Problems?

In some optimization problems, you *can* have slack in the inequality (i.e., $\lambda_1 < p_1 - p_2$).

Example where it doesn't bind:

Suppose we had a different utility function (not perfect substitutes), and at the optimum:

- $x_1 = 0$ (still at the boundary)
- But switching to $x_1 = -\epsilon$ wouldn't actually help much (maybe the utility loss is too high)

In that case, λ_1 could be less than $p_1 - p_2$.

The Key Difference

For **perfect substitutes**: goods are exactly equivalent for utility, so switching between them is purely about prices. This makes the shadow price exactly equal to the price difference.

For **other utility functions**: goods are not perfect substitutes, so the shadow price depends on marginal utilities, not just prices.

7 Practical Advice for Problem 2b

What to Write on Your Homework

Approach 1 (Simpler): Just set $\lambda_1 = p_1 - p_2$ and verify it satisfies all conditions.

Approach 2 (More rigorous):

1. Derive that $\lambda_1 \leq p_1 - p_2$
2. Use economic reasoning: the shadow price of $x_1 \geq 0$ equals the price difference
3. Conclude $\lambda_1 = p_1 - p_2$
4. Verify $\lambda_1 \geq 0$ requires $p_1 \geq p_2$

Either approach is fine for this problem!

8 Summary

The Bottom Line

Why does $\lambda_1 = p_1 - p_2$ instead of $\lambda_1 < p_1 - p_2$?

1. For perfect substitutes, the corner solution is unique and fully determined
2. The shadow price λ_1 measures the cost savings from relaxing $x_1 \geq 0$
3. This cost savings is exactly $p_1 - p_2$ (the price difference)
4. Therefore λ_1 takes its maximum feasible value
5. Mathematically: the inequality allows $\lambda_1 \in [0, p_1 - p_2]$, but the economics pins down $\lambda_1 = p_1 - p_2$

In general: Not all inequalities in KKT conditions bind. But for this specific problem with perfect substitutes, it does.

9 Alternative Perspective: Solving Directly

Actually, there's another way to see this that avoids the question entirely.

We have 5 unknowns: $x_1, x_2, \lambda_1, \lambda_2, \lambda_3$

From our guess and complementary slackness:

$$x_1 = 0 \quad (\text{given}) \tag{3}$$

$$x_2 = u^* \quad (\text{from utility constraint}) \tag{4}$$

$$\lambda_2 = 0 \quad (\text{from complementary slackness}) \tag{5}$$

$$\lambda_3 = p_2 \quad (\text{from FOC for } x_2) \tag{6}$$

That's 4 equations. We have one more FOC inequality:

$$-p_1 + \lambda_1 + \lambda_3 \leq 0$$

Substituting $\lambda_3 = p_2$:

$$\lambda_1 \leq p_1 - p_2$$

Now, we want to find THE optimal solution. Since the problem has a unique minimum (perfect substitutes have unique corner solutions), all the variables should be uniquely determined.

The only way λ_1 is uniquely determined is if the inequality binds: $\lambda_1 = p_1 - p_2$.

If λ_1 could be anything in $[0, p_1 - p_2]$, we wouldn't have a unique solution to the KKT system, which contradicts the fact that our original optimization problem has a unique solution.

Uniqueness Argument

The original problem (minimize cost subject to constraints) has a unique solution because perfect substitutes always lead to corner solutions.

Therefore, the KKT system must also have a unique solution.

Therefore, all variables (including λ_1) must be uniquely determined.

Therefore, the inequality must bind: $\lambda_1 = p_1 - p_2$.